

Overview

Patient Retention Rate 60.0%	Total Patients 5	Retained Patients 3
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Executive Summary

This dashboard provides insights into **patient retention** and **marketing effectiveness** for prescription renewals.

Key Business Questions:

- Patient Retention:** What percentage of patients renew their prescriptions, and how does this trend over time?
- Marketing Attribution:** Which marketing touchpoints drive prescription renewals?

Navigation:

- Patient Retention Analysis** – Detailed monthly retention trends
- Marketing Attribution** – Campaign and channel performance analysis

Patient Retention Analysis

Understanding Patient Retention

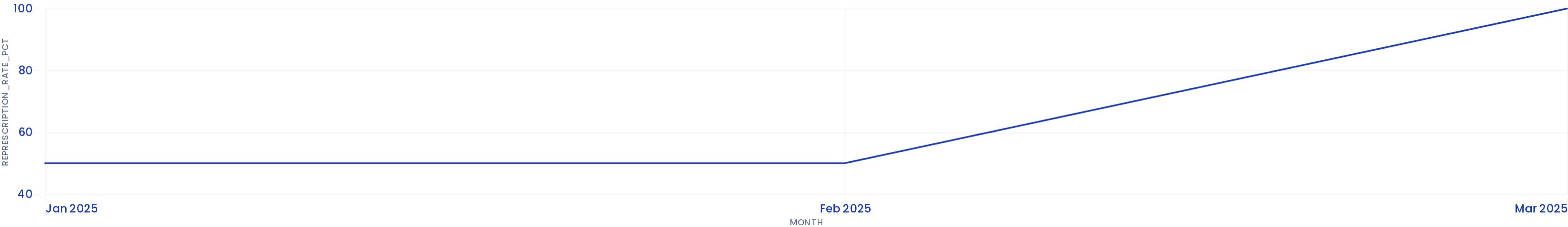
Business Context: Patient retention measures how many patients continue their prescribed treatment by getting re-prescriptions. High retention indicates effective treatment outcomes and patient satisfaction.

Metrics:

- **Retention Rate** = (Patients who renewed ÷ Total patients) × 100%
- Calculated at the **patient level** (not prescription level) to avoid double-counting
- Tracked monthly to identify seasonal trends and program effectiveness

Monthly Patient Retention Rate

Percentage of patients who renewed their prescription each month



Retention Performance

Total Patients	Retained Patients	Retention Rate %
5	3	60.0%

Marketing Attribution

Business Question

"Which touchpoint contributes the most to re-prescription?"

Analytical Approach: Last-Touch Attribution

Methodology

We use a **last-touch attribution model** that credits the final marketing interaction before a patient renews their prescription:

1.

Identify renewal events:

Select all prescriptions where `represcription = TRUE`
2.

Find eligible touchpoints:

For each renewal, identify all touchpoints between `prescription_start` and `represcription_date`
3.

Attribute to last touchpoint:

Use window function to rank touchpoints by date (descending) and credit the most recent one
4.

Aggregate results:

Count distinct patients attributed to each touchpoint type and calculate percentage

SQL Logic

```
ROW_NUMBER() OVER (
  PARTITION BY patient_id, prescription_id
  ORDER BY touchpoint_date DESC  -- Last touch!
) = 1
```

Why Last-Touch?

- Identifies the **immediate conversion driver** before renewal
- Answers: "What was the final nudge that drove the patient to renew?"
- Common in performance marketing to optimize conversion tactics

Key Findings

Based on current data:

- **CHECK_IN_CALL** (66.7%) – Phone calls are the dominant last touchpoint
- **EMAIL_NUDGE** (33.3%) – Email reminders also drive conversions
- Average timing: 65–75 days after initial prescription

Limitations & Caveats

1. Correlation ≠ Causation

Issue: The last touchpoint may have occurred *after* the patient decided to renew.

- A patient might schedule a renewal, then receive a routine check-in call
- The call didn't cause the renewal—it just happened to be last

2. Recency Bias

Issue: Over-credits the final interaction, ignores earlier influential touchpoints.

- Example: An educational email campaign (first-touch) might have been more influential than the final reminder call, but gets zero credit

3. Selection Bias

Issue: Only analyzes patients who renewed—no comparison group.

- We don't know: Did non-renewed patients receive different touchpoints?
- Without this, we can't say touchpoints *caused* renewals

4. Missing Confounders

Issue: Doesn't control for patient characteristics that drive renewal.

- Factors not measured: disease severity, medication effectiveness, doctor relationship, insurance coverage, socioeconomic status
- High-risk patients may naturally renew regardless of touchpoints

5. Multi-Touch Journey Ignored

Issue: Patients typically interact multiple times across channels.

- Attributing 100% credit to one touchpoint oversimplifies the customer journey
- Alternative models (linear, time-decay, position-based) distribute credit more fairly

6. No Counterfactual

Issue: We don't know what would have happened without touchpoints.

- Would patients have renewed anyway?
- Requires A/B testing: randomly assign patients to "touchpoint" vs. "no touchpoint" groups

Recommendations for Improvement

Immediate (use existing data)

1.

Compare with first-touch attribution

(already in `@2_gold.touchpoint_attribution_first_touch`)
2.

Analyze non-renewals:

Which touchpoints did patients who *didn't* renew receive?
3.

Segment by patient characteristics:

Does attribution differ by age, insurance type, or doctor?

Medium-term (additional analysis)

1.

Multi-touch attribution:

Test time-decay or position-based models
2.

Propensity score matching:

Match similar patients who renewed vs. didn't renew, control for confounders
3.

Survival analysis:

Model time-to-renewal with touchpoints as time-varying covariates

Long-term (experimentation)

1.

Randomized controlled trial:

Randomly assign patients to receive/not receive specific touchpoints
2.

Holdout testing:

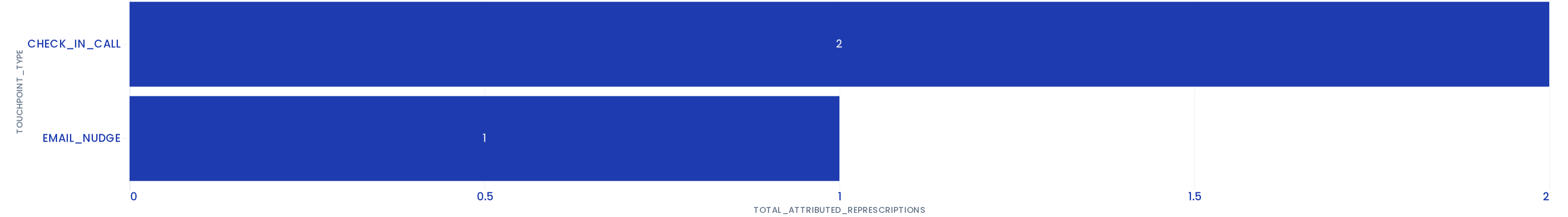
Create control groups that receive no marketing touchpoints
3.

Incrementality testing:

Measure the causal lift of each touchpoint type

Attributed Renewals by Touchpoint Type

Number of patient renewals attributed to each touchpoint type



Detailed Attribution Breakdown

Touchpoint	Channel	Outcome	Renewals	Attribution %	Avg Days to Renewal
CHECK_IN_CALL	Phone	Completed	2	66.7%	65
EMAIL_NUDGE	Email	Clicked	1	33.3%	75